

A Global Occupation Conjecture for the Accelerated $3x + 1$ Map: Divergent-Orbit Sparsity, Exact Realizers, and the Moving-Anchor Problem

Elias De Jesús
Independent Researcher
ORCID: 0009-0007-0190-9143
dejesuselias10@gmail.com

Revision 5, 2026

Abstract

We study the Effective Occupation Conjecture (EOC) for the accelerated $3x+1$ map $T(m) = (3m+1)/2^{a(m)}$, $a(m) = \nu_2(3m+1)$. Writing $R_n = S_n - n \log_2 3$ for the drift of the accumulated 2-adic valuation and $O_c(m_0) = \#\{n \geq 0 : R_n \leq c\}$ for the occupation of a fixed corridor $c > 0$, EOC asserts $O_c(m_0) = O(\log m_0)$ for every seed m_0 and every c . On the divergence side, we incorporate the strongest unconditional result currently available: building on a 1999 collision-free window theorem of Garcia and Tal and an explicit quantitative sharpening supplied by Curry, every divergent (equivalently, injective) accelerated orbit has summable reciprocals, $\sum_n 1/m_n < \infty$, and consequently satisfies no eventual logarithmic drift floor $R_n \geq -B \log_2 n + O(1)$ for any $B < 1/\beta_* \approx 1.0358567$. This crosses the harmonic barrier $B = 1$ and improves on our elementary bound $B < 8/9$, obtained by a one-step fixed-modulus congruence-packing argument; we show that this specific mechanism reaches its terminal exponent at $8/9$. We then give an exact residue-axis reformulation: realizers of a finite valuation word occupy a single residue class modulo 2^{S_N+1} , with an explicit corrected formula, and EOC's single-window form is equivalent to an exponential lower bound on least positive realizers. Such bounds are proved for periodic and eventually periodic symbolic sectors, together with a first-defect bound controlled by the periodic run preceding the defect, through a fixed 2-adic anchor mechanism. The corresponding problem remains open for genuinely irregular words, where the relevant 2-adic anchor moves with the word. As a further application of the exact realizer machinery, we prove and formally verify in Lean that every finite binary history of a natural mod-32 return observable associated with the pattern $(2, 1)$ is realizable by a genuine accelerated $3x+1$ orbit, with the prescribed events occurring consecutively. This finite universality supplies no asymptotic equidistribution result; rather, it rules out finite forbidden-pattern mechanisms as a route to pointwise balance. The entropy deficit $I_{\text{Collatz}} = 0.0793186 \dots$ calibrates the natural population scale but, by itself, supplies no pointwise anti-concentration; finite computation is consistent with this distinction but does not prove it. Neither EOC nor the Collatz conjecture is proved here. The resulting picture separates substantial unconditional control of hypothetical divergent orbits from the unresolved pointwise arithmetic-placement problem for exact realizers, which remains the central obstacle to EOC.

Non-claims. This paper proves neither the Collatz conjecture nor any tier of EOC. It does not prove an exponential realizer floor for arbitrary confined words, and it does not promote any computational finding to a theorem. All conjectural statements are labeled as such throughout.

Contents

1	Introduction	2
2	Preliminaries	3
3	Layer I: EOC and its logical status	4
3.1	Four tiers	4
3.2	Logical consequences, and why effective constants matter	4
3.3	EOC as an occupation-time shadow of Rozier’s hypothesis	5
4	Layer II: unconditional drift-axis progress	5
4.1	An elementary $8/9$ bound	6
4.2	The Garcia–Tal/Curry sparsity bound and the $1/\beta_*$ exclusion	7
5	Layer III: the residue-axis reformulation	8
5.1	The exact realizer congruence	8
5.2	Confinement and the single-window equivalence	9
5.3	Proved sectors	10
5.4	Finite universality of a mod-32 return observable	11
6	Computational calibration	14
7	Relation to prior work	15
8	Open problems	15
9	Conclusion	16
A	Band Rigidity	17
B	Compact status summary	18
C	Further Clarifications on the Finite Return Construction	18

1 Introduction

The accelerated $3x + 1$ map sends an odd positive integer m to the next odd term of its Collatz trajectory, $T(m) = (3m + 1)/2^{a(m)}$, $a(m) = \nu_2(3m + 1)$. Writing S_n for the accumulated 2-adic valuation after n steps and $R_n = S_n - n \log_2 3$ for its drift below the critical slope $\log_2 3$, the Effective Occupation Conjecture (EOC) asserts that the time an orbit spends with R_n at or below any fixed level is logarithmic in the seed: this single pointwise statement, if true, forces every orbit to be eventually periodic and reduces the Collatz conjecture to a finite cycle search (Theorem 3.6).

This revision is organized around three layers, reflecting where the project now stands.

Layer I (§3) states EOC precisely, in four tiers of increasing strength, and proves what each tier implies for Collatz.

Layer II (§4) treats the drift (Archimedean) axis: what is unconditionally known about how negative R_n can be forced to run, for a hypothetical divergent orbit. This axis has changed materially since the previous revision. We give our own elementary bound $B < 8/9$ in brief, together with the precise sense in which it is terminal for its method, and then present the substantially stronger

unconditional result now available through Garcia and Tal's 1999 collision-free window theorem together with an explicit quantitative form recently supplied by Curry: $B < 1/\beta_* \approx 1.0358567$, crossing the harmonic barrier $B = 1$.

Layer III (§5) treats the residue (arithmetic-placement) axis: the exact congruence governing which integers realize a given valuation word, and the equivalence between EOC's single-window form and an exponential floor on least realizers. This is now the conceptual center of the paper. Beyond the periodic and eventually periodic realizer-floor results, the exact realizer machinery also supports a sharp finite-history universality statement for a concrete return observable related to Chang's mod-16/mod-32 one-bit orbit-mixing classification (§5.4): the machinery can prescribe arbitrary finite labeled histories of this observable. This finite universality is a structural consequence of the realizer congruence, not a step toward asymptotic equidistribution; it shows only that finite forbidden-pattern arguments cannot settle the corresponding pointwise balance question.

The central message of this revision is a single distinction that organizes everything below:

Divergent-orbit sparsity is now considerably better understood, through a mechanism external to and structurally different from our own drift-axis argument. EOC itself remains open, because its pointwise residue-axis core — the placement of exact realizers of genuinely irregular confined words — is untouched by this advance.

Garcia–Tal/Curry's result controls nothing about hypothetical nontrivial cycles, the finite occupation of convergent orbits, arbitrary confined-word least realizers, the moving-anchor anti-concentration problem, or effective EOC constants. We return to this boundary explicitly through-out.

2 Preliminaries

Let m_0 be a positive odd integer. The accelerated odd map is

$$T(m) = \frac{3m+1}{2^{a(m)}}, \quad a(m) = \nu_2(3m+1) \geq 1,$$

sending positive odd integers to positive odd integers. Write $m_{n+1} = T(m_n)$, $d_n = a(m_n)$, $S_n = \sum_{i < n} d_i$, $\alpha = \log_2 3$, $R_n = S_n - n\alpha$. Each step changes R by $d_n - \alpha \in \{1 - \alpha, 2 - \alpha, \dots\}$, and $R_0 = 0$.

Remark 2.1. If every accelerated orbit reaches 1, the Collatz conjecture holds: every positive integer's odd part has an accelerated orbit, and the classical trajectory of an odd number interleaves the accelerated iterates with halvings. We work exclusively with T .

For fixed $c > 0$ define the occupation functional $O_c(m_0) = \#\{n \geq 0 : R_n \leq c\} \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$. This is a global sojourn count, distinct from stopping time: it neither halts at first passage nor counts time far above the critical line.

Lemma 2.2 (Eliahou–Rozier identity). *For every odd $m_0 \geq 1$ and every $n \geq 0$,*

$$m_n 2^{R_n} = m_0 \prod_{i < n} \left(1 + \frac{1}{3m_i}\right), \quad \text{equivalently} \quad R_n = \log_2 \frac{m_0}{m_n} + E_n, \quad E_n = \sum_{i < n} \log_2 \left(1 + \frac{1}{3m_i}\right) \geq 0.$$

Proof. $m_{i+1} = \frac{3m_i}{2^{d_i}} \left(1 + \frac{1}{3m_i}\right)$; multiply over $i < n$ and use $3^n 2^{-S_n} = 2^{-R_n}$. □

This identity is due to Eliahou [1] (cycles) and Rozier [2] (general trajectories); we claim no novelty for it.

Lemma 2.3 (Wedge). *If $m_i \geq m_0$ for all $0 \leq i \leq n$, then $R_i \leq E_i \leq i/(3m_0 \ln 2)$ for all $0 \leq i \leq n$. In particular $R_i \leq c$ for $0 \leq i \leq \min\{n, \lfloor 3c \ln 2 \cdot m_0 \rfloor\}$.*

Proof. For $m_k \geq m_0$, $\log_2(1 + \frac{1}{3m_k}) = \frac{\ln(1 + 1/(3m_k))}{\ln 2} \leq \frac{1}{3m_k \ln 2} \leq \frac{1}{3m_0 \ln 2}$; sum over $k < i$ and apply Lemma 2.2. $\square$

3 Layer I: EOC and its logical status

3.1 Four tiers

Conjecture 3.1 (Existence EOC). *For every fixed $c > 0$ there exist constants K_c, C_c such that every positive odd m_0 satisfies $O_c(m_0) \leq K_c \log_2 m_0 + C_c$.*

Conjecture 3.2 (Effective EOC at (c, K, C)). *For stated $c > 0$ and stated numerical K, C , every positive odd m_0 satisfies $O_c(m_0) \leq K \log_2 m_0 + C$.*

Conjecture 3.3 (Sharp-leading EOC). *For every fixed $c > 0$, $O_c(m_0) \leq K \log_2 m_0 + O_c(\log \log m_0)$, $K = 1/I(\alpha)$, where $I(\mu) = \mu - \mu \log_2 \mu + (\mu - 1) \log_2(\mu - 1)$.*

Conjecture 3.4 (Strong sharp EOC). *For every fixed $c > 0$ there is C_c with $O_c(m_0) \leq K \log_2 m_0 + C_c$.*

Conjecture 3.4 $\Rightarrow$ 3.3 $\Rightarrow$ 3.1; Conjecture 3.2 at any constants implies Conjecture 3.1 at that c . Crucially, Conjecture 3.1 does not imply Conjecture 3.2 at any stated constants, since it asserts existence without a bound — this distinction drives the whole of §3.2. All four tiers are pointwise statements about every seed, not almost-every-seed statements.

3.2 Logical consequences, and why effective constants matter

Definition 3.5. For $c > 0$, constants K, C , let $m_* = m_*(K, C, c)$ be least such that $3c \ln 2 \cdot m > K \log_2 m + C + 1$ for all $m \geq m_*$; it exists and is computable by finitely many exact comparisons.

Theorem 3.6. *Fix $c > 0$ and suppose Conjecture 3.1 holds at (c, K, C) ; put $m_* = m_*(K, C, c)$. Then:*

- (i) *No accelerated orbit is injective (Definition 4.1); every orbit is eventually periodic.*
- (ii) *Every odd $m_0 \geq m_*$ has an iterate $m_i < m_0$ with $i \leq \lfloor 3c \ln 2 \cdot m_0 \rfloor$.*
- (iii) *If some seed's orbit does not reach 1, there is a nontrivial accelerated cycle with minimum member below m_* .*
- (iv) *If additionally no nontrivial cycle has minimum member below m_* , the Collatz conjecture holds.*

Proof. (i) By Corollary 4.6 below, an injective orbit has $O_c(m_0) = \infty$, contradicting the finite bound. (ii) Let $H = \lfloor 3c \ln 2 \cdot m_0 \rfloor$; if $m_i \geq m_0$ for $i \leq H$ then Lemma 2.3 gives $R_i \leq c$ throughout, so $O_c(m_0) \geq H + 1$, exceeding $K \log_2 m_0 + C$ by the defining property of m_* — contradiction. (iii) By (i) the orbit is eventually periodic in a nontrivial cycle; every iterate of its minimum member $m_{\min}$ lies on the cycle, hence $\geq m_{\min}$, so if $m_{\min} \geq m_*$, (ii) gives a smaller iterate on the cycle, contradiction. (iv) Combine (iii) with the added hypothesis. $\square$

Corollary 3.7. *Assume Conjecture 3.1 at (c, K, C) . Then Collatz is equivalent to: no nontrivial accelerated cycle has minimum member below $m_*(K, C, c)$.*

Remark 3.8 (Why effective constants are genuinely needed). m_* is computable from (K, C, c) , but Conjecture 3.1 supplies no bound on (K, C) themselves, hence none on m_* . One cannot repair this by minimal-counterexample reasoning: Theorem 3.6(ii) localizes a counterexample below m_* but does not eliminate it. A nontrivial cycle would cost enormously — since Barina’s verification [13] settles all seeds below 2^{71} , any nontrivial cycle’s minimum member exceeds 2^{71} , and the wedge (Lemma 2.3) applied to that member forces $O_1(m_{\min}) \gtrsim 3 \ln 2 \cdot 2^{71} \approx 4.9 \times 10^{21}$ — a value Conjecture 3.1 can certainly accommodate at unspecified constants, but which is wildly larger than anything the record table of §6 suggests (max observed ratio under 10). This inconsistency-with-computation is not a consequence of Conjecture 3.1 itself; it is exactly why the effective tier matters. Furthermore, the affine identity $\nu_2(3(4m+1)+1) = 2 + \nu_2(3m+1)$, $T(4m+1) = T(m)$, shows the set of non-convergent seeds is closed under $m \mapsto 4m+1$ if nonempty — but such seeds descend immediately ($T(4m+1) \leq (3m+1)/2 < 4m+1$), voiding the wedge at the first step, so no amplification argument removes the base range $m < m_*$; only effective constants or an independent cycle exclusion below m_* can.

Corollary 3.9 (Explicit instance). *Assume Conjecture 3.2 at $c = 1$, $K = 13$, $C = 100$. Then $m_*(13, 100, 1) = 90$; every odd $m < 90$ reaches 1 by direct inspection (longest: $m = 27$, 41 accelerated steps); hence Collatz holds.*

Proof. $f(m) = 3 \ln 2 \cdot m - 13 \log_2 m - 101$ is increasing for $m \geq 10$ and $f(89) < 0 < f(90)$. $\square$

Remark 3.10. $K = 13$ exceeds the predicted asymptotic slope $1/I(\alpha) = 12.6074\dots$; $C = 100$ exceeds by two orders of magnitude the maximum ratio observed computationally (§6). m_* is insensitive to the constants: at $K = 100, C = 10^6$ one still finds $m_* < 5 \times 10^5$, inside the published verification range.

3.3 EOC as an occupation-time shadow of Rozier’s hypothesis

Theorem 3.11. *Assume Rozier’s Lower Bound Hypothesis [2, Hyp. 2.3]. Then for every fixed $c > 0$, $O_c(m_0) \leq \frac{1}{I(\alpha)} \log_2 m_0 + O_c(\log \log m_0)$, i.e. LBH implies Conjecture 3.3, hence Conjecture 3.1.*

The proof applies Rozier’s own maneuver [2, Thm. 4.1] at occupation times rather than at total stopping time, via the entropy identity $\mu(1 - H(1/\mu)) = I(\mu)$; we regard it as a corollary of his framework, recorded here because the occupation functional had not previously been isolated. It does not advance the provability of EOC — LBH is itself open, and arguably harder — but positions EOC precisely as a weaker occupation-time shadow of LBH. Making the additive term in Conjecture 3.3 explicit would supply Conjecture 3.2 and, with it, Collatz via Corollary 3.9’s route, given verification below the resulting m_* ; this is Open Problem A.

4 Layer II: unconditional drift-axis progress

This section is unconditional throughout; nothing here assumes EOC.

Definition 4.1. An orbit (m_n) is injective if $m_0, m_1, m_2, \dots$ are pairwise distinct.

Lemma 4.2 (Terminal and cyclic drift). *If an orbit reaches 1, or enters a nontrivial positive cycle, then $R_N \rightarrow +\infty$. Consequently every non-injective orbit has $R_N \rightarrow +\infty$.*

Proof. Both cases are instances of the orbit entering a periodic cycle of odd states, length $L \geq 1$, total valuation S over one period. Cycle closure means L applications of T return to the starting value m ; the identity $2^S m = 3^L m + C_L(D)$ (Proposition 5.2's proof, with $C_L(D) > 0$ a sum of positive terms) gives $m(2^S - 3^L) = C_L(D)$, forcing $2^S > 3^L$: strictly positive drift per period. Writing $N = k + jL + i$ for the orbit's entry index k and $0 \leq i < L$, $R_N \geq R_k + j(S - L\alpha) - \Delta \rightarrow +\infty$ as $j \rightarrow \infty$, where Δ is the (fixed, cycle-dependent) maximal within-period excursion. A non-injective orbit repeats some state, hence is eventually periodic from that point, placing it in this case. $\square$

4.1 An elementary 8/9 bound

Lemma 4.3 (Carry budget). *Every accelerated iterate m_k , $k \geq 1$, satisfies $m_k \not\equiv 0 \pmod{3}$, since $3m + 1 \equiv 1 \pmod{3}$ and division by 2^a is multiplication by a unit mod 3. Hence for $k \geq 1$ the states lie in the density-1/3 set of integers coprime to 6. For every injective orbit and every $N \geq 1$, $E_N \leq \frac{1}{9} \log_2 N + O(1)$, hence $2^{E_N} \ll N^{1/9}$.*

Proof sketch. The map $m \mapsto \log_2(1 + \frac{1}{3m})$ is decreasing, so E_N is maximized by the extremal packing of the N smallest integers coprime to 6 (the term at $k = 0$ contributes a bounded $O(1)$). Since these enumerate as the interleaved progressions $6k + 1, 6k + 5$, their harmonic sum is $\frac{1}{3} \ln N + O(1)$ by direct partial summation; the claim follows from $\log_2(1 + x) \leq x / \ln 2$. The unweighted packing over merely-odd states gives the weaker $\frac{1}{6} \log_2 N + O(1)$; the mod-3 input is a single elementary sharpening of this specific fixed-modulus mechanism. $\square$

Lemma 4.4 (Carry demand). *With $U_k = 2^{E_k}$, $U_{k+1} - U_k = 2^{R_k} / (3m_0)$ exactly.*

Proof. $2^{E_k} - 1 = (1 + \frac{1}{3m_k}) - 1 = \frac{1}{3m_k}$ exactly, so $U_{k+1} - U_k = 2^{E_k} / (3m_k)$; by Lemma 2.2 in exponentiated form, $2^{R_k} = (m_0/m_k)2^{E_k}$. $\square$

Theorem 4.5 (Logarithmic-floor exclusion, 8/9). *Let m_0 have an injective accelerated orbit. For every $B < 8/9$, $R_k \geq -B \log_2 k$ fails for infinitely many k .*

Proof. If $R_k \geq -B \log_2 k$ held for $k \geq N_0$, telescoping Lemma 4.4 gives $\sum_{N_0 \leq k < N} k^{-B} \ll_{m_0} U_N \ll N^{1/9}$ by Lemma 4.3; but the left side grows like $N^{1-B}/(1-B)$, incompatible with $N^{1/9}$ once $1 - B > 1/9$. $\square$

Corollary 4.6. *For every injective orbit and fixed $c > 0$: $\#\{0 \leq j < N : R_j > c\} \ll_{m_0, c} N^{1/9}$, hence $O_c(m_0) = \infty$ at density-one rate, and Conjecture 3.1 directly excludes injective orbits.*

Remark 4.7 (Terminality of the mod-3 mechanism). For a prime $p > 3$, both 2 and 3 are units mod p , so $3m + 1 \equiv 0 \pmod{p}$ is solvable and no annihilation analogous to the mod-3 case occurs: this specific one-step fixed-modulus packing mechanism supplies no further density reduction at any other fixed modulus (an exhaustive scan below 2×10^5 found no unvisited residue class at any of several tested moduli, consistent with, though not proving, this). Pushing past 8/9 toward 1 therefore requires either a genuinely different mechanism, or a graded harmonic-thinness statement $\sum_{k < N} 1/m_k \leq (\frac{1}{3} - \eta) \ln N + O(1)$ for some $\eta > 0$. We record this not as an open problem — §4.2 resolves it, more strongly than requested — but to make precise exactly what kind of external input was missing, and hence exactly why the following result does not weaken Theorem 4.5's own terminality claim: it succeeds by a mechanism of the different kind this remark anticipates.

4.2 The Garcia–Tal/Curry sparsity bound and the $1/\beta_*$ exclusion

Garcia and Tal proved in 1999, for Hasse’s generalized $3n+1$ -type maps (a family that includes the classical map), that every infinite orbit has Banach density zero [5]. The mechanism is a collision-freeness observation: if an orbit is infinite and aperiodic, no two of its values can ever coincide under further iteration — coincidence would force eventual periodicity — so the orbit’s value set embeds into a set of representatives of density zero. This qualitative result carried no explicit rate. Curry [6] supplies an explicit quantitative form of the same mechanism, applied as a windowed pigeonhole argument over parity prefixes (in contrast to Theorem 4.5’s one-step, fixed-modulus argument):

Theorem 4.8 (Garcia–Tal [5]; explicit form, Curry [6]). *Let $\mathcal{O}$ be the value set of an infinite aperiodic orbit of the raw map $T_0(n) = n/2$ (n even), $(3n+1)/2$ (n odd). For every $\beta > \beta_*$ there is C_β with*

$$\#(\mathcal{O} \cap [a, a+X)) \leq C_\beta X^\beta \log(2X) \quad \text{uniformly in } a,$$

where $\beta_* = \gamma_* \log_2 3 \approx 0.9653844$ and $\gamma_* \in (\frac{1}{2}, \frac{1}{\log_2 3})$ is the unique root of the entropy-crossing equation $H(\gamma) = \gamma \log_2 3$ ($\gamma_* \approx 0.6090897$, H the binary entropy).

We do not reproduce Curry’s proof, which pigeonholes each window by the number of odd steps in a length- N parity prefix ($N \approx \log_2 X$), splits into a heavy case (many odd steps, bounded by an entropy tail $X^{H(\gamma)}$) and a light case (few odd steps, where Lemma 2.2-type contraction forces a collision within $X^{\gamma \log_2 3}$ values, exploiting collision-freeness directly), and optimizes γ at the crossing where the two bounds balance. The constant γ_* coincides with Rozier’s critical ones-ratio r_H [2], arising from the same entropy-versus-contraction balance.

We now derive, self-containedly, what this gives for the accelerated map, in our own notation, translating via $g_n := n\alpha - S_n = -R_n$ (Curry’s sign convention).

Proposition 4.9 (Reciprocal summability). *Every divergent (equivalently, injective, by Lemma 4.2) orbit satisfies $\sum_{x \in \mathcal{O}} 1/x < \infty$, and hence, for its accelerated odd iterates, $\sum_{n \geq 0} 1/m_n < \infty$.*

Proof. Divergence makes the orbit aperiodic, hence collision-free by the argument above. Fix $\beta \in (\beta_*, 1)$ (nonempty since $\beta_* < 1$) and sum Theorem 4.8 over dyadic blocks:

$$\sum_{x \in \mathcal{O}} \frac{1}{x} = \sum_{i \geq 0} \sum_{x \in \mathcal{O} \cap [2^i, 2^{i+1})} \frac{1}{x} \leq \sum_{i \geq 0} C_\beta 2^{i\beta} (i+2) \log 2 \cdot 2^{-i} = C_\beta \log 2 \sum_{i \geq 0} (i+2) 2^{-i(1-\beta)} < \infty,$$

since $\beta < 1$. The accelerated (odd) subsequence $\{m_n\} \subseteq \mathcal{O}$, so $\sum_n 1/m_n$ is a sub-sum of a convergent sum of positive terms. $\square$

This is the graded harmonic-thinness input identified in Remark 4.7, in a strictly stronger form: the sum is bounded, not merely $(\frac{1}{3} - \eta) \ln N + O(1)$. In the carry-budget notation of §4, $E_N = \sum_{k < N} \log_2(1 + \frac{1}{3m_k})$ is bounded along every divergent orbit, rather than of size $\frac{1}{9} \log_2 N$.

Theorem 4.10 (Floor exclusion through the harmonic barrier). *No divergent orbit satisfies $R_n \geq -B \log_2 n + O(1)$ eventually, for any $B \leq 1$.*

Proof. By Lemma 2.2, $m_n = m_0 2^{-R_n} Q_n$, $Q_n = \prod_{j < n} (1 + \frac{1}{3m_j})$ nondecreasing. Since $\log(1+x) \leq x$, Proposition 4.9 gives $\log Q_n = \sum_{j < n} \log(1 + \frac{1}{3m_j}) \leq \frac{1}{3} \sum_j 1/m_j < \infty$, so $Q_n \uparrow Q_\infty < \infty$. If $R_n \geq -B \log_2 n - K$ eventually, then $2^{-R_n} \leq 2^{K+nB}$, so $m_n \leq m_0 Q_\infty 2^{K+nB}$, giving $1/m_n \geq cn^{-B}$; then $\sum 1/m_n \geq c \sum n^{-B} = \infty$ for $B \leq 1$, contradicting Proposition 4.9. $\square$

Theorem 4.11 (Occupation bound and the $1/\beta_*$ threshold). *Fix $\beta \in (\beta_*, 1)$. There are constants c_1, c_2 , depending on β and on the orbit through m_0, Q_∞ , with*

$$\#\{n \geq 0 : R_n \geq -G\} \leq c_1 2^{\beta G} (G + c_2) \quad \text{for all } G \geq 1.$$

Consequently, no divergent orbit satisfies $R_n \geq -B \log_2 n + O(1)$ eventually, for any

$$B < \frac{1}{\beta_*} \approx 1.0358567 \dots$$

The endpoint $B = 1/\beta_$ is not excluded by this argument.*

Proof. If $R_n \geq -G$ then $2^{-R_n} \leq 2^G$, so $m_n \leq m_0 Q_\infty 2^G$ by the identity above; since the m_n are pairwise distinct elements of the collision-free set $\mathcal{O}$, apply Theorem 4.8 with $a = 1$, $X = m_0 Q_\infty 2^G$ to bound their count, giving the stated form. Now suppose $R_n \geq -B \log_2 n - K$ eventually with $B > 0$ (the case $B \leq 0$ is Theorem 4.10). For every $n \leq 2^{(G-K)/B}$ beyond the eventual threshold, $-B \log_2 n - K \geq -G$, so such n all satisfy $R_n \geq -G$, giving $\#\{n : R_n \geq -G\} \geq \frac{1}{2} 2^{(G-K)/B}$ for large G . Comparing with the occupation bound and taking logarithms forces $B \geq 1/\beta$ for every $\beta > \beta_*$; letting $\beta \rightarrow \beta_*^+$ gives $B \geq 1/\beta_*$. The contrapositive is the stated exclusion. $\square$

Remark 4.12. In the drift notation used throughout this paper, Theorem 4.11 reads: no injective accelerated orbit satisfies $R_k \geq -B \log_2 k$ eventually, for any $B < 1.0358567 \dots$, exceeding both the harmonic barrier $B = 1$ and Theorem 4.5's bound $B < 8/9$. This does not weaken Theorem 4.5 or Remark 4.7: the one-step, fixed-modulus congruence-packing mechanism remains terminal at $8/9$ for arguments of that kind; Theorem 4.8's mechanism is a windowed argument over a modulus scaling like 2^N , using entropy counting and collision-freeness directly, genuinely different in kind. We keep Theorem 4.5 in this paper because it is elementary, accelerated-map-native, and independent of external input, and because it identifies precisely the point at which a structurally different mechanism becomes necessary — which Theorem 4.11 then supplies.

Remark 4.13 (Scope). Nothing in this section constrains hypothetical nontrivial cycles (a periodic orbit is not aperiodic, hence outside every hypothesis above), proves any tier of EOC, or touches the occupation of convergent orbits. Divergent-orbit sparsity and EOC's pointwise residue-axis core (§5) are logically independent problems; this section resolves the first far more completely than the paper's earlier drift-axis argument alone could, and leaves the second exactly as open.

5 Layer III: the residue-axis reformulation

This is the actual unresolved core of EOC. Divergence-side control bounds how sparse an orbit's near-critical returns can be; it says nothing about which integers realize a given near-critical valuation pattern, or how small such a realizer can be. That is a 2-adic residue-placement question, entirely orthogonal to the Archimedean sparsity question of §4.

5.1 The exact realizer congruence

Definition 5.1. A valuation word $D = (d_0, \dots, d_{N-1})$ has $d_i \geq 1$; $s_0 = 0$, $s_j = \sum_{i < j} d_i$, $S_N(D) = s_N$, and carry sum $C_N(D) = \sum_{j=0}^{N-1} 3^{N-1-j} 2^{s_j}$. An odd m_0 realizes D if $a(m_i) = d_i$ for all $i < N$ along its orbit.

Proposition 5.2 (Exact realizer congruence). *The exact realizers of D form a single residue class modulo $2^{S_N(D)+1}$, not $2^{S_N(D)}$. Writing $r(D)$ for the least positive representative,*

$$r(D) \equiv (2^{S_N(D)} - C_N(D)) \cdot 3^{-N} \pmod{2^{S_N(D)+1}}.$$

Proof. From $2^{d_j}m_{j+1} = 3m_j + 1$, dividing by 2^{s_j} and telescoping gives $2^{s_j}m_j = 3^j m_0 + q_j$, $q_0 = 0$, $q_{j+1} = 3q_j + 2^{s_j}$ (so $q_N = C_N(D)$).

(a) *Integrality.* If m_0 satisfies the aggregate congruence $2^{S_N} \mid 3^N m_0 + C_N(D)$, downward induction on j shows every formal iterate is an integer: given $2^{s_{j+1}} \mid 3^{j+1}m_0 + q_{j+1}$, the recursion gives $3^{j+1}m_0 + q_{j+1} = 3(3^j m_0 + q_j) + 2^{s_j}$; since $2^{s_j} \mid 2^{s_{j+1}}$ divides the left side and the explicit 2^{s_j} term, it divides their difference $3(3^j m_0 + q_j)$, and as 3 is a unit mod any power of 2, $2^{s_j} \mid 3^j m_0 + q_j$.

(b) *Interior oddness.* With all m_j integers, $2^{d_j}m_{j+1} = 3m_j + 1$ holds exactly; since $d_j \geq 1$ the left side is even, so m_j is odd for every $j < N$.

(c) *Exactness.* For $j < N - 1$ both m_j, m_{j+1} are odd, so $\nu_2(3m_j + 1) = d_j$ automatically. At $j = N - 1$, exactness requires additionally that m_N be odd.

(d) *Terminal parity.* “ m_N odd” is equivalent to $2^{S_N}m_N \equiv 2^{S_N} \pmod{2^{S_N+1}}$ (a standard fact about odd multiples), i.e. (using $2^{S_N}m_N = 3^N m_0 + C_N(D)$ identically) to $3^N m_0 + C_N(D) \equiv 2^{S_N} \pmod{2^{S_N+1}}$; solving for m_0 using that 3^N is a unit mod 2^{S_N+1} gives the stated congruence. $\square$

Remark 5.3 (Sanity check). For $D = (1)$: the coarse congruence gives only $m \equiv 1 \pmod{2}$; the exact congruence gives $m \equiv 3 \pmod{4}$. Indeed $m = 3$ realizes $D = (1)$ ($\nu_2(10) = 1$) while $m = 1$ does not ($\nu_2(4) = 2$). Verified against direct orbit iteration on hundreds of random cases with zero discrepancies.

Definition 5.4 (Coarse anchor vs. exact lift). For a word D of length N and any $S \leq S_N(D)$, the coarse anchor at depth S is $a_S(D) := (-C_N(D) \cdot 3^{-N}) \bmod 2^S \in [0, 2^S)$, using the aggregate congruence alone. Proposition 5.2 shows $r(D)$ reduces mod 2^{S_N} to $a_{S_N}(D)$, and is one of the two candidates $\{a_{S_N}(D), a_{S_N}(D) + 2^{S_N}\}$, with terminal parity selecting which. In particular, unconditionally, $r(D) \geq a_{S_N}(D)$: this one inequality, not the equality (which need not hold), is what every floor argument below actually uses.

5.2 Confinement and the single-window equivalence

Definition 5.5. Fix $c > 0$. D (length N) is c -confined if $R_j(D) \leq c$ for all $0 \leq j \leq N$. For odd m_0 , $L_c(m_0) = \max\{N \geq 0 : \text{the length-}N \text{ prefix of } m_0\text{'s word is } c\text{-confined}\}$, finite by Lemma 4.2. Always $O_c(m_0) \geq L_c(m_0)$. For the c -confined family at depth N , $r_{\min}(N, c) := \min_D r(D)$.

Proposition 5.6 (Record-chronology inversion). $r_{\min}(N, c) = \min\{m \text{ odd} : L_c(m) \geq N\}$.

Proof. If $r(D) = m$ for c -confined D of length N , m 's own word matches D , so $L_c(m) \geq N$; conversely if $L_c(m) \geq N$, m 's own length- N prefix is c -confined and realized by m . $\square$

This inversion turns the abstract minimization over words into a direct seed scan, exploited computationally in §6.

Proposition 5.7 (Single-window equivalence). Fix $c > 0$. For the same $\varepsilon > 0$ (any $\varepsilon \leq 2$), the following are equivalent:

- (i) $r(D) \geq 2^{\varepsilon N}$ for every c -confined D of length N with $r(D) \geq 3$;
- (ii) $L_c(m_0) \leq \varepsilon^{-1} \log_2 m_0$ for every odd $m_0 \geq 3$.

Proof sketch. $r(D) = 1$ occurs only for $D = (2, \dots, 2)$ (the accelerated fixed point at 1), which is why both statements exclude the small values $r(D) < 3$, $m_0 < 3$. (i) $\Rightarrow$ (ii): put $m_0 = r(D_L)$ for D_L the length- $L = L_c(m_0)$ prefix of m_0 's own orbit; if $r(D_L) \geq 3$, (i) gives $m_0 \geq 2^{\varepsilon L}$ directly, while the excluded case $r(D_L) = 1$ forces L bounded by a constant depending only on c . (ii) $\Rightarrow$ (i): for c -confined D with $r(D) \geq 3$, put $m_0 = r(D)$; then $L_c(m_0) \geq N$, and (ii) gives $r(D) \geq 2^{\varepsilon N}$. $\square$

Equivalently, (ii) is the target exponential floor $r_{\min}(N, c) \geq 2^{\varepsilon N} - O(1)$. This is the exact remaining pointwise core of EOC's single-window form — not a new object, but a precise reformulation as a statement about where the least realizer of a confined word can lie. The multi-window conjecture (Conjecture 3.1) is strictly stronger: it implies (ii) $\Leftrightarrow$ (i), with no converse in general, since L_c constrains only a word's initial confined run while O_c counts every later return (witnessed by, e.g., $m_0 = 285175$ with $L_1 = 14$ but $O_1 = 97$).

5.3 Proved sectors

For a fixed block X ($q_X = |X|$, $S_X = S(X)$, $\delta_X = 3^{q_X} - 2^{S_X}$), the quantity $\xi_X = -C(X)/\delta_X \in \mathbb{Z}_2$ is well-defined since δ_X is odd. Unwinding the aggregate congruence through the exact carry recursion for X^n shows the coarse anchor satisfies $a_{nS_X}(X^n) = \xi_X \bmod 2^{nS_X}$ for every n : realizer classes of X^n reduce, at depth nS_X , to successive truncations of one fixed 2-adic anchor.

Theorem 5.8 (Periodic realizer floor). *Suppose $\xi_X \notin \mathbb{Z}_{>0}$ (the excluded case contains genuine periodic realizations such as $X = (2)$, for which $\xi_X = 1$, $r(X^n) = 1$ for every n). Then there is a finite B_X with $\xi_X \bmod 2^{nS_X} \geq 2^{nS_X - B_X}$ for all sufficiently large n ; combined with $r(X^n) \geq a_{nS_X}(X^n)$, this gives $r(X^n) \geq 2^{nS_X - B_X}$, hence $\log_2 r(X^n)/(nS_X) \rightarrow 1$.*

The mechanism: ξ_X 's binary expansion is eventually periodic (period $B_X = \text{ord}_{b_X}(2)$ for the reduced odd denominator b_X), and eventual periodicity of a non-positive-integer rational forces every sufficiently deep truncation to carry a 1-bit within bounded distance of its top. The terminal-parity correction never weakens this inequality, only possibly strengthens it. Full proof: companion note [15].

Theorem 5.9 (Eventually periodic words). *For a fixed finite prefix Y , the same mechanism applies to YX^n with transformed anchor $\xi_{X,Y} = -C(Y)3^{-q_Y} + 2^{S_Y}3^{-q_Y}\xi_X$; outside the same sign-refined exceptional case, eventual periodicity of $\xi_{X,Y}$ gives a finite $B_{X,Y}$ with $r(YX^n) \geq 2^{S_Y + nS_X - B_{X,Y}}$, hence $\log_2 r(YX^n)/(S_Y + nS_X) \rightarrow 1$.*

A fixed finite prefix changes only the constant $B_{X,Y}$; it cannot change the asymptotic exponent. Full proof: [15].

Proposition 5.10 (Bulk-defect anchor stability). *For $D = X^a EX^b$ (fixed X , arbitrary finite defect E , arbitrary b), the coarse anchor satisfies $a_{aS_X}(D) = \xi_X \bmod 2^{aS_X}$ exactly, independent of E and b . Consequently, outside the exceptional case, $r(D) \geq 2^{aS_X - B_X}$ for a sufficiently large.*

This bound depends only on a , the length of the periodic run before the first defect; it is unaffected by what follows.

Remark 5.11 (Retraction). An earlier draft of this program conjectured that a long periodic recovery run after a defect restores global anchor precision for sparse, growing-gap defects. This is false: the correct carry-concatenation composition is nested from the suffix inward, with the first block outermost, not the last, so later recovery does not improve Proposition 5.10's bound. The surviving, correct statement is exactly Proposition 5.10.

Remark 5.12 (Arbitrary irregular words: open). For a general confined word, the coarse anchor $\xi_N(D) = -C_N(D)3^{-N}$ changes with N in a way tied to the word's own non-repeating letters: there is no single fixed rational number to invoke periodicity of, because there is no single number. This is the genuine unresolved sector — the moving-anchor sector — and it is exactly where actual Collatz orbits live: record-holding confined seeds (e.g. $m_0 = 27$, confined for 39 of its 41 steps) have no long repeated block, and independently record-holding seeds' own valuation words differ

in over half their positions at matched length, ruling out a shared-structure explanation. A natural compressed statistic for this sector, the pairwise collision law $\nu_2(C(W_1) - C(W_2)) = S(P) + \min(a, b)$ for equal-length words sharing prefix P and first differing letters a, b , was investigated and found to reduce to ordinary symbolic prefix geometry, with no residual arithmetic content. We record this as a genuine, checked negative result. This is Open Problem D, the paper's central open problem.

5.4 Finite universality of a mod-32 return observable

The exact realizer congruence of §5.1 does more than pin down the residue class of a given word: it can be used to *prescribe* arbitrary finite histories of a concrete arithmetic return observable, as we now show. The observable in question is closely related to the burst-ending, mod-32 classification isolated by Chang [17, 18] as a concrete one-bit reformulation of the pointwise obstruction in his framework. We state the connection to Chang's work carefully, since the attribution has two distinct parts.

Chang works with the compressed odd-to-odd map, isolates the dominant burst-ending channel $n \equiv 1 \pmod{8}$, and shows [17, §5] that within this channel the outcome of a burst-to-gap transition depends on exactly one bit: $n \equiv 9 \pmod{32}$ corresponds to a gap of length ≥ 2 , and $n \equiv 25 \pmod{32}$ corresponds to a unit gap. He formulates the remaining obstruction as a pointwise balance question between these two mod-32 classes along the burst-ending subsequence of an individual orbit, and shows that the map itself is balanced (his Map Balance Theorem), so any residual bias is orbit-level. That attribution — the burst-ending observable, the dominant $n \equiv 1 \pmod{8}$ channel, the 9-versus-25 mod-32 split, and the pointwise balance obstruction — is Chang's.

What follows below — the valuation-word reformulation of this observable, the $(2, 1)$ event criterion, the canonical block construction, and the finite-history realizability theorem — is ours, built from the exact realizer machinery of §5.1 above. We have not located an explicit statement of the valuation-word criterion in Chang's supplied papers in this exact form; we derive it here directly rather than attribute it.

Definition 5.13. For a valuation word $D = (d_0, d_1, \dots)$ with realizer $m = r(D)$ and orbit (m_j) , we say a *Chang event* occurs at index j (with $j + 1 < N$) if

$$m_j \equiv 9 \pmod{16}.$$

For such an orbit, direct calculation gives the equivalent valuation criterion

$$m_j \equiv 9 \pmod{16} \iff (d_j, d_{j+1}) = (2, 1).$$

At a Chang event,

$$m_j \equiv 9 \pmod{32} \iff d_{j+2} = 1, \quad m_j \equiv 25 \pmod{32} \iff d_{j+2} \geq 2.$$

Thus Chang's two mod-32 classes correspond respectively to the valuation patterns $(2, 1, 1)$ and $(2, 1, \geq 2)$.

For $K \geq 1$ and $y = (y_0, \dots, y_{K-1}) \in \{0, 1\}^K$, define the *canonical Chang word* $D(y)$ by concatenating, for $t = 0, \dots, K - 1$, the block

$$\beta(y_t) = (2, 1, 1) \text{ if } y_t = 0, \quad \beta(y_t) = (2, 1, 2) \text{ if } y_t = 1,$$

so that

$$D(y) = \beta(y_0)\beta(y_1)\cdots\beta(y_{K-1})$$

has length $3K$.

Theorem 5.14 (Finite Chang-history realizability). *Let $K \geq 1$ and $y = (y_0, \dots, y_{K-1}) \in \{0, 1\}^K$, and let $m = r(D(y))$ be the least positive realizer of $D(y)$ furnished by Proposition 5.2. Then:*

(i) *For every j with $j + 1 < 3K$, a Chang event occurs at j if and only if*

$$j \in \{0, 3, 6, \dots, 3(K-1)\};$$

there is no Chang event at any other j with $j + 1 < 3K$.

(ii) *For each $t = 0, \dots, K-1$,*

$$m_{3t} \equiv 25 \pmod{32} \iff y_t = 1, \quad m_{3t} \equiv 9 \pmod{32} \iff y_t = 0.$$

(iii)

$$1 \leq m < 2^{4K + \#\{t: y_t=1\}+1}.$$

The event status is completely determined through $j = 3K-2$. The final prescribed event occurs at $j = 3K-3$, while $j = 3K-2$ is controlled and is not an event. The first event position not determined by the length- $3K$ canonical word is $j = 3K-1$, since testing the event criterion there requires the next valuation digit d_{3K} , which lies outside $D(y)$.

Proof. Within each block $\beta(y_t)$, the three digits are

$$(d_{3t}, d_{3t+1}, d_{3t+2}) = (2, 1, y_t+1).$$

At $j = 3t$,

$$(d_{3t}, d_{3t+1}) = (2, 1),$$

so a Chang event occurs. Moreover,

$$d_{3t+2} = 1 \iff y_t = 0, \quad d_{3t+2} \geq 2 \iff y_t = 1.$$

By Definition 5.13, this gives

$$m_{3t} \equiv 9 \pmod{32} \iff y_t = 0, \quad m_{3t} \equiv 25 \pmod{32} \iff y_t = 1,$$

proving (ii).

For (i), it remains to check that no spurious event occurs at $j = 3t+1$ or $j = 3t+2$ within the controlled range $j+1 < 3K$.

At $j = 3t+1$,

$$(d_{3t+1}, d_{3t+2}) = (1, y_t+1) \in \{(1, 1), (1, 2)\},$$

so the pair is never $(2, 1)$.

At $j = 3t+2$, this index lies in the controlled range only when $t < K-1$, and then

$$(d_{3t+2}, d_{3t+3}) = (y_t+1, 2) \in \{(1, 2), (2, 2)\},$$

again never $(2, 1)$.

Hence the only indices j with $j+1 < 3K$ for which

$$(d_j, d_{j+1}) = (2, 1)$$

are

$$j = 0, 3, \dots, 3(K-1).$$

By Definition 5.13, these are exactly the Chang-event positions, proving (i).

For (iii), by Definition 5.1,

$$S_{3K}(D(y)) = \sum_{t < K} (2 + 1 + (y_t + 1)) = 4K + \#\{t : y_t = 1\},$$

since each block contributes 4 when $y_t = 0$ and 5 when $y_t = 1$. By Proposition 5.2, the least positive realizer $m = r(D(y))$ satisfies

$$1 \leq m < 2^{S_{3K}(D(y))+1},$$

and therefore

$$1 \leq m < 2^{4K + \#\{t: y_t = 1\} + 1}.$$

This is a direct instantiation of the exact realizer congruence at $N = 3K$; no additional estimate is introduced. $\square$

Corollary 5.15. *There are no forbidden finite binary Chang histories: for every $K \geq 1$, every $y \in \{0, 1\}^K$ is realized, in the sense of Theorem 5.14, by some positive odd seed.*

This corollary should not be over-read. It does not imply that arbitrary *infinite* binary histories are realizable, that the associated symbolic system is a full shift, or that any asymptotic equidistribution statement follows. The theorem is a finite statement: for each finite K , a specific finite construction realizes y , with a seed bound growing like $2^{4K+O(K)}$. Nothing here addresses what happens along a single infinite orbit's own, uncontrolled sequence of Chang events.

Remark 5.16 (Confinement of the canonical family). It is natural to ask whether the canonical words of Definition 5.13 lie in the c -confined family of Definition 5.5, since that family organizes the rest of §5. The answer is mixed, and depends on y .

For the all-zero history ($y_t = 0$ for all t , block $(2, 1, 1)$ repeated), the within-block drift increments are $2 - \alpha$, $1 - \alpha$, $1 - \alpha$, and the running maximum of R_j over the whole word is attained at $j = 1$, where $R_1 = 2 - \alpha \approx 0.4150375$ (the net per-block drift $4 - 3\alpha \approx -0.7551$ is negative, so later blocks start from an ever-lower baseline and never exceed this first-step peak). Hence the all-zero canonical family is c -confined for $c \geq 2 - \alpha \approx 0.4150375$, in particular for the corridor $c = 1$ used elsewhere in this paper — but *not* for $0 \leq c < 2 - \alpha$.

For the all-one history (block $(2, 1, 2)$ repeated), the net per-block drift is $5 - 3\alpha \approx +0.2451$, strictly positive, so $R_{3K} \rightarrow +\infty$ as $K \rightarrow \infty$: sufficiently long all-one canonical words are not c -confined for any fixed c .

The general lesson is that finite Chang-history universality (Theorem 5.14) is not the same statement as universality inside the c -confined EOC family of §5: some canonical histories are confined for relevant c , and others are decisively not. The canonical construction should not be read as occupying, still less resolving, the genuinely irregular confined-word sector of Remark 5.12.

Remark 5.17 (What this does and does not do). Arbitrarily long finite biased runs of either label are realizable, across suitable seeds, by Theorem 5.14. Consequently, no proof of asymptotic balance for Chang's return observable can rest on excluding a finite forbidden pattern, on local exclusion of one of the two labels, or on exclusion of long finite biased runs: all of these occur, for suitable (and explicitly constructed) seeds. This sharpens rather than removes the obstruction Chang identifies: the finite arithmetic is maximally permissive, and the remaining question is genuinely asymptotic and dynamical, concerning the behavior of Chang events along a single infinite orbit's own deterministic sequence, not the existence of any finite pattern.

A short formal-verification note: the finite realizability statement of Theorem 5.14 has been formalized and machine-checked in Lean 4/Mathlib in the module `EOC/ChangHistory.lean` of the public EOC verification repository, compiling with zero `sorry/admit` and included in the normal `lake build`. We do not describe the standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`) as defects. The Lean theorem names `chang_history_labels`, `changWord_event_positions`, `no_internal_chang_event`, and `finite_chang_history_realizable` correspond respectively to the label identity of part (ii), the event-position claim of part (i), the no-spurious-event half of part (i), and the existence statement underlying Corollary 5.15.

6 Computational calibration

This section reports, compactly, the computational findings that sharpen the interpretation of Open Problem D without resolving it. Full data and methodology are in the companion notes [15, 16]; here we retain only what materially constrains the search for a pointwise mechanism.

Entropy scale. At critical density $\rho = \log_3 2$, the number of length- N near-critical valuation words grows like $2^{\alpha H_2(\rho)N + o(N)}$ (Terras–Everett [3, 4]), while the ambient residue space at typical valuation $S \approx \alpha N$ has size $2^{\alpha N}$. The resulting deficit,

$$I_{\text{Collatz}} = 1 - \frac{H_2(\rho)}{\rho} = 0.0793186127748554\dots \text{ per odd step,}$$

calibrates the natural scale of the problem but is a population-level statement: it does not, by itself, constrain where in the residue space any particular word’s realizer lies.

Shell injectivity. For fixed (N, S) , the map $D \mapsto \xi_N(D) \bmod 2^S$ is injective across all $\binom{S-1}{N-1}$ compositions of S into N parts (a corollary of the collision law of Remark 5.12). The shell therefore contains exactly $\binom{S-1}{N-1}$ distinct residues among 2^S available positions — ordinary counting, no collision-driven reduction.

No second exponential rate. Exhaustive enumeration of confined words across several near-critical shells (S up to 30, shell size up to 3.5×10^7) measured the endpoint depth $E(D) = S - \log_2 r(D)$ against the random-injective null (uniform sampling of the same population size without replacement). Across all tested shells and thresholds, the observed tail count stayed within a factor 1.0–1.2 of the null prediction, with no trend toward either 0 or ∞ ; the extremal gap $H_2(\rho)S - \max_D E(D)$ stayed bounded rather than growing. Matched comparison between confined and unconfined words at equal (N, S) found no detectable deformation attributable to confinement itself. Separately, balanced (near-equal-letter) words, tested to total valuation depth $S = 50,000$, showed no systematic realizer suppression relative to generic words. Conclusion: bulk entropy, shell counting, confinement, and ordinary extreme-value statistics are all consistent with the admissible residues behaving like a random injective placement at the population size entropy predicts; none of these mechanisms shows evidence of a second exponential rate. Any successful pointwise argument must detect arithmetic information finer than all four.

Global record frontier. Using Proposition 5.6, a direct seed scan (rather than word enumeration, which becomes misleading once residue pinning is accounted for) tracked confinement-length records $L_1(m)$ through 1.25×10^9 odd seeds, recovering champions including 27, 703, 60975, 837799, 6649279, $\dots$ up to a record $L_1 = 252$ at $m = 156977263$, with no further record through the full scanned range. Comparing $\log_2 m_k$ against $I_{\text{Collatz}} L_k$ for the mature record range ($65 \leq L_k \leq 252$) shows a bounded additive defect (≈ 7 –10 bits, essentially flat), consistent with, but not proof of, $\log_2 r_{\min}(N, 1) = I_{\text{Collatz}} N + O(1)$. This is an empirical observation over a finite range, not a theorem.

Remark 6.1 (The finite-history theorem is a different kind of evidence). Theorem 5.14 is exact and deterministic: it produces, for each prescribed finite history, a specific integer with a specific (bounded, explicit) size, and does not rely on any of the random-injective null models used elsewhere in this section. It should not be read alongside the entropy-calibration and record-frontier material above as though it were evidence of the same kind; it is a structural consequence of Proposition 5.2 applied to a specific family of words, not a computational or statistical finding, and it supplies no new entropy rate, no rate-function interpretation, and no lower bound on least realizers beyond the elementary upper bound already used in its own proof.

7 Relation to prior work

Rozier [2] is the closest neighbor: the product identity, the harmonic distinct-values packing, the entropy calibration, and the implication method used in Theorem 3.11 are his or Eliahou’s [1]. This paper’s own contributions are the occupation functional itself, its pointwise Collatz reduction (Theorem 3.6), the corrected residue-axis reformulation (Proposition 5.2), the periodic/eventually-periodic realizer floors (§5), and the finite Chang-history realizability theorem (§5.4), formally verified in Lean. Terras and Everett [3, 4] supply the parity-vector correspondence underlying the exact realizer construction. Garcia and Tal [5] supply the collision-free window mechanism underlying §4.2’s qualitative content; Curry [6] supplies its explicit quantitative form, the exponent β_* , and the resulting threshold $1/\beta_*$ — the strongest currently available unconditional divergence-side result, and we attribute it precisely as such: not an EOC theorem, but a substantially stronger elementary drift-axis bound than §4.1’s own. Lagarias–Weiss [7] and Kontorovich–Lagarias [8] supply the stochastic rate framework of which $I(\alpha)$ is a member, with functionals (first passage, extremal excursion) distinct from occupation. Tao [9] controls minimum values attained in logarithmic density; Open Problem F asks whether a density-one EOC is accessible by his methods. The cycle-exclusion literature [10, 11, 12] and Barina’s verification [13] are what Corollary 3.9 consumes; we prove nothing new about cycles. The Sturmian combinatorics of Morse–Hedlund [14] underlie the rigid-sector result of Appendix A.

For §5.4: the burst-ending observable, the dominant $n \equiv 1 \pmod{8}$ channel, the 9-versus-25 mod-32 classification, and the pointwise-balance obstruction are Chang’s [17], whose companion paper [18] develops the broader structural framework, spectator-bit behavior, and the distributional-to-pointwise obstruction his program shares with ours. The valuation-word reformulation, the (2, 1) event criterion, the canonical block construction, and Theorem 5.14 itself are derived here from the exact realizer machinery of §5.1; we have not verified that this exact reformulation appears in Chang’s supplied papers, and attribute it to the present work accordingly.

8 Open problems

Open Problem A (Effective EOC constants). Establish Conjecture 3.2 at any explicit (c, K, C) , or any explicit upper bound on admissible constants in Conjecture 3.1. This closes the Collatz implication outright (Corollary 3.9).

Open Problem B (Cycle exclusion below threshold). Show no nontrivial accelerated cycle has minimum member below $m_*(K, C, c)$. Settled for $m_* \leq 2^{71}$ by [13]; open beyond.

Open Problem C (Arbitrary-word exponential realizer floor). Find $\varepsilon > 0$ with $r(D) \geq 2^{\varepsilon N} - O(1)$ for all genuinely relevant c -confined words, not merely periodic, eventually periodic, or first-defect words. This is the residue-axis core of EOC (Proposition 5.7).

Open Problem D (Moving-anchor anti-concentration). Identify a deterministic arithmetic property of $-C_N(D)3^{-N} \pmod{2^{S_N}}$ that prevents exceptionally small realizers in arbitrary irregular confined words. §6’s computations show this cannot come from bulk entropy, shell counting, confinement, or ordinary extreme-value behavior; it must be a genuinely pointwise arithmetic or Dio-phantine mechanism. This is the paper’s central open problem, and Open Problems C–D are, in substance, the same question at two levels of precision.

Open Problem E (Frontier-defect control). A weaker target than Open Problem D, suggested but not established by §6: $E(D) \leq H_2(\rho_c)S + \Phi(N)$ for every c -confined D , with $\Phi(N) = o(N)$. Even $\Phi(N) = O(\log N)$ would be valuable. Whether this is a genuinely easier target than Open Problem C, or logically equivalent to it, is itself open.

Open Problem F (Density-one EOC). Can a density-one form of EOC be proved by the methods of Tao [9]? This remains meaningful after §4.2: divergent-orbit sparsity says nothing about the occupation of orbits that do converge, which is what a density-one EOC statement would address.

Open Problem G (Pointwise Chang return balance). For an infinite aperiodic accelerated orbit, let z_t denote successive visits to the event class $n \equiv 9 \pmod{16}$ (Definition 5.13), and put $Y_t = \mathbf{1}[z_t \equiv 25 \pmod{32}]$.

A natural strong form asks whether

$$\#\{t < K : Y_t = 1\}/K \longrightarrow 1/2.$$

Chang’s own reduction [17] requires only the weaker quantitative condition that the imbalance $|\#\{t \leq m : Y_t = 1\}/\#\{t \leq m : n_{t_i} \equiv 9 \text{ or } 25\} - 1/2|$ remain within a tolerance $\delta < \delta_{\max}$, for a $\delta_{\max}$ determined by his block-total-variation budget — not exact convergence to $1/2$.

Theorem 5.14 shows that neither the strong form nor Chang’s own weaker quantitative form can be settled merely by excluding finite binary histories: every finite history occurs, for suitable seeds. We do not claim this problem is equivalent to EOC; it is a related but distinct return-distribution problem, arising from a different construction (the compressed odd-to-odd map’s burst structure) than the drift-axis occupation functional O_c studied elsewhere in this paper.

9 Conclusion

We have stated EOC in four tiers and proved precisely what each implies for Collatz. On the drift axis, we retain an elementary, self-contained $8/9$ logarithmic-floor exclusion and show that this exponent is terminal for the specific one-step fixed-modulus congruence-packing mechanism developed here. We then present the substantially stronger unconditional result now available externally: building on Garcia and Tal’s 1999 collision-free window theorem, Curry’s quantitative sharpening gives reciprocal summability along every divergent orbit and a logarithmic-floor exclusion reaching $B < 1/\beta_* \approx 1.0358567$. This crosses the harmonic barrier $B = 1$ by a windowed entropy-and-collision mechanism structurally different from the fixed-modulus argument.

On the residue axis, we give the corrected exact realizer congruence and show that EOC’s single-window form is equivalent to an exponential lower bound on least positive realizers of confined valuation words. Such bounds are proved for periodic and eventually periodic symbolic sectors through a fixed 2-adic anchor mechanism, together with a first-defect bound controlled by the periodic run preceding the defect. These arguments do not extend to genuinely irregular words, for which the relevant 2-adic anchor moves with the word itself. Computation calibrates this moving-anchor sector against the known entropy deficit and is consistent with random-injective placement at the corresponding population scale, but supplies no pointwise theorem.

As a further, independent application of the exact realizer machinery, we showed that it can prescribe arbitrary finite histories of a concrete mod-32 return observable related to Chang’s one-bit orbit-mixing reduction: the realizer machinery now does more than identify the residue class of a valuation word, it can prescribe arbitrary finite histories of a nontrivial arithmetic return observable. Yet this finite universality sharpens rather than removes the corresponding obstruction: since every finite history is realizable, any proof of asymptotic balance for that observable must come from genuinely long-range, dynamical control of a single orbit’s own Chang-event sequence, rather than from finite symbolic exclusion. This is the same moving-anchor theme that organizes the rest of the paper, instantiated on a second, independent return observable; we do not claim that resolving it is necessary for EOC, or that it moves EOC’s own central Open Problem D any closer to resolution.

The resulting separation is sharp: substantial unconditional information is now available about the sparsity and drift behavior of hypothetical divergent orbits, while EOC itself remains open because these results do not control the arithmetic placement of individual exact realizers. The central unresolved problem is therefore pointwise: to identify a deterministic arithmetic or Diophantine mechanism that prevents exceptionally small realizers for arbitrary irregular confined words. Neither EOC nor the Collatz conjecture is proved here.

Acknowledgments

AI-assisted tools, including Anthropic’s Claude, were used for adversarial auditing, computational verification, algebra checking, and drafting, including an independent audit of Curry’s argument reported to have found no substantive error, and including drafting assistance in preparing the finite Chang-history material of §5.4. Lean 4/Mathlib was used to machine-check the finite Chang-history formalization (`EOC/ChangHistory.lean`) in the public EOC verification repository; the formalization compiles without `sorry` or `admit` and is included in the normal build. We do not describe this machine-checking as an independent philosophical proof; it is a statement that the formalized proof compiles under Lean’s standard axioms. All research direction, interpretation, final mathematical judgments, and responsibility for the manuscript remain with the author.

A Band Rigidity

Theorem A.1 (Band Rigidity). *Let $\theta = \alpha - 1$. If a word’s drift path satisfies $\beta \leq R_j < \beta + 1$ for all $j \leq \ell$ and some $\beta \in \mathbb{R}$, then every letter is in $\{1, 2\}$, and (writing $b_i = d_i - 1$) the word is a length- ℓ factor of the Sturmian system of slope θ ; conversely every such factor arises, and there are exactly $\ell + 1$ admissible words for each ℓ .*

Proof sketch. A step with $d_i \geq 3$ changes R by $\geq 3 - \alpha > 1$, leaving any unit band, forcing $d_i \in \{1, 2\}$. Confinement to a unit band forces $B_j := S_j - j = \lceil j\theta + \beta \rceil$ for all $j \leq \ell$; the resulting b_j is the upper mechanical word of slope θ , and by Morse–Hedlund [14] these are exactly the length- ℓ Sturmian factors, $\ell + 1$ in number. $\square$

By Corollary 4.6, the rigid corridor is pointwise excluded as an eventual home for divergence candidates; the theorem delimits exactly where deterministic control is available (rigid, zero-entropy sector) versus where only density statements hold (entropic sector). Occupation record orbits hover loosely, with longest width-one runs of length 5–7 inside occupation windows of length 41–357.

B Compact status summary

Statement	Status	Note
Exact realizer congruence (Prop. 5.2)	Proved	corrects an earlier draft’s residue
8/9 floor exclusion (Thm. 4.5)	Proved	elementary, terminal for its mechanism
$1/\beta_*$ floor exclusion (Thm. 4.11)	External theorem	Garcia–Tal [5]; Curry [6]
Reciprocal summability (Prop. 4.9)	Proved	here, from Thm. 4.8 via dyadic blocks
Periodic/eventually-periodic realizer floors (§5)	Proved	full proofs in [15]
Finite Chang-history realizability (Thm. 5.14)	Proved + Lean verified	every finite binary history occurs as consecutive events; finite statement only
Shell injectivity, entropy scale (§6)	Proved known	/ population-level only
No second exponential shell rate	Computational	not a theorem
Existence EOC and all tiers	Conjectural	
Arbitrary-word realizer floor (OP C–D)	Open	the residue-axis core
No cycle below m_* (OP B)	Open	settled for $m_* \leq 2^{71}$
Pointwise Chang return balance (OP G)	Open	finite universality gives no asymptotic frequency theorem

C Further Clarifications on the Finite Return Construction

This appendix records several technical and interpretive clarifications concerning the finite-history construction of §5.4. It introduces no new theorem and does not strengthen Theorem 5.14. Its purpose is to make explicit the boundary between what is controlled by a finite valuation word and what remains an asymptotic property of a single orbit.

The mod-32 return observable motivating the construction comes from Chang’s one-bit orbit-mixing framework [17, 18], whereas the finite valuation-word construction and its exact realizability statement are derived in the present paper from the exact realizer congruence of Proposition 5.2.

A. Exact endpoint of the controlled event window

Let

$$D(y) = \beta(y_0)\beta(y_1)\cdots\beta(y_{K-1}), \quad \beta(0) = (2, 1, 1), \quad \beta(1) = (2, 1, 2),$$

so that $D(y)$ has length $3K$. If $m = r(D(y))$ is its least positive realizer, let $(m_j)_{j \geq 0}$ denote the full accelerated orbit of m . The construction controls the valuation digits

$$d_0, d_1, \dots, d_{3K-1},$$

but it does not prescribe d_{3K} .

As used in Definition 5.13, within the controlled valuation prefix the event criterion is

$$m_j \equiv 9 \pmod{16} \iff (d_j, d_{j+1}) = (2, 1).$$

Therefore the event status at index j is determined whenever both d_j and d_{j+1} belong to the prescribed word, namely whenever

$$j + 1 < 3K.$$

Equivalently, the event status is completely determined for

$$0 \leq j \leq 3K - 2.$$

Within this controlled range, Theorem 5.14 shows that the only event positions are

$$j = 0, 3, 6, \dots, 3(K - 1) = 3K - 3.$$

In particular, the final prescribed event at $3K - 3$ is fully controlled.

The next index, $j = 3K - 2$, is also controlled and is not an event, because

$$(d_{3K-2}, d_{3K-1}) = (1, y_{K-1} + 1) \neq (2, 1).$$

The first index whose event status is not determined by the canonical word is therefore

$$j = 3K - 1.$$

Determining whether an event occurs there would require the pair

$$(d_{3K-1}, d_{3K}),$$

and d_{3K} lies outside the prescribed length- $3K$ valuation word.

Thus the exact endpoint is

The event status is completely controlled through $j = 3K - 2$. The first uncontrolled event position is $j = 3K - 1$.

No statement concerning an event at $j = 3K - 1$ or later is required for Theorem 5.14.

B. The orbit continues beyond the prescribed word

The notation in Theorem 5.14 should not be interpreted as terminating the orbit after $3K$ valuation digits. The realizer $m = r(D(y))$ is an ordinary positive odd integer and therefore determines a full accelerated orbit

$$m_0, m_1, m_2, \dots$$

What terminates after $3K$ digits is only the part of the valuation sequence prescribed by the canonical construction. Thus one should distinguish

$$\underbrace{(m_j)_{j \geq 0}}_{\text{full genuine orbit}} \quad \text{from} \quad \underbrace{(d_0, \dots, d_{3K-1})}_{\text{controlled valuation prefix}}.$$

Proposition 5.2 guarantees that this prescribed finite valuation word is realized by an actual positive odd seed. This finite-prefix viewpoint is consistent with the classical parity-vector correspondence underlying accelerated Collatz coding [3, 4].

Theorem 5.14 therefore makes an exact statement about the events determined by the controlled prefix. It makes no assertion about the continuation of the event sequence after that prefix.

C. Full finite-history language does not imply a full shift

Corollary 5.15 establishes that for every $K \geq 1$, all 2^K binary histories occur across suitable seeds. Equivalently, the full binary finite-history language is realizable length by length.

This does not imply that every infinite binary sequence occurs along a single accelerated orbit. In particular, it does not establish a full-shift structure, a symbolic conjugacy, topological entropy $\log 2$, Bernoulli behavior, independence of successive labels, or any limiting frequency law.

The quantifiers are fundamentally different. Theorem 5.14 supplies

$$\forall K \geq 1 \forall y \in \{0, 1\}^K \exists m = m(K, y)$$

such that the prescribed finite history occurs.

This does not imply

$$\forall (y_t)_{t \geq 0} \exists m \forall K \geq 1$$

such that the first K event labels of the same orbit equal

$$y_0, \dots, y_{K-1}.$$

The seed in the finite theorem is allowed to depend on both K and the prescribed history. Passing from finite realizability to an infinite realization would require an additional compatibility or compactness argument not provided here.

D. What finite universality rules out

Although finite universality gives no asymptotic frequency theorem, it does eliminate a broad class of possible finite obstructions.

For every K , both extreme histories

$$(0, 0, \dots, 0) \in \{0, 1\}^K \quad \text{and} \quad (1, 1, \dots, 1) \in \{0, 1\}^K$$

are realizable by suitable seeds. More generally, every mixed binary pattern of length K is realizable by Theorem 5.14.

Therefore an asymptotic balance argument cannot succeed merely by showing that

- one of the two labels is locally forbidden;
- some fixed finite binary word cannot occur;
- runs of one label have a universal finite maximum length; or
- sufficiently long finite biased histories are arithmetically impossible.

Theorem 5.14 rules out each such mechanism at finite depth. Any proof of pointwise balance must instead exploit information that persists across arbitrarily long portions of one fixed orbit.

This is a negative structural conclusion about possible proof mechanisms, not a positive proof of balance. In particular, it proves neither the stronger exact-equipartition formulation of Open Problem G nor the weaker sufficient-balance condition appearing in Chang's reduction [17].

E. Finite prescription and pointwise asymptotics

The exact realizer congruence is highly effective for finite prescription. Once a positive valuation word D is specified, Proposition 5.2 identifies a genuine residue class of seeds realizing that word. The canonical construction of Definition 5.13 exploits this fact by encoding each desired binary label into a short valuation block.

Accordingly, the finite problem has the form

prescribe a word $\longrightarrow$ solve an exact residue condition $\longrightarrow$ obtain a genuine finite orbit prefix.

The asymptotic problem has a different logical direction. Starting from one fixed seed, the future valuation word is no longer chosen freely. One must instead understand the deterministic sequence generated by that seed:

$$m_0 \longrightarrow d_0, d_1, d_2, \dots \longrightarrow Y_0, Y_1, Y_2, \dots$$

Thus Theorem 5.14 demonstrates arithmetic freedom across the population of seeds, whereas Open Problem G asks for long-run statistical control along an individual orbit.

The former does not imply the latter. This distinction parallels the broader distributional-to-pointwise obstruction emphasized in Chang’s framework [17, 18], although the finite realizability theorem itself is specific to the construction developed in the present paper.

F. Relation to the moving-anchor problem

There is a useful conceptual analogy between the finite-history result and the moving-anchor obstruction of §5, but no equivalence is claimed.

For EOC, the unresolved question concerns the arithmetic placement of the exact realizer of an irregular confined valuation word. Periodic and eventually periodic words can be controlled because their coarse realizers arise from truncations of a fixed 2-adic anchor; the corresponding fixed-anchor realizer floors are developed in [15].

For genuinely irregular words, the relevant anchor changes with the word, and the fixed-rational periodicity argument disappears, as described in Remark 5.12.

The finite-history problem exhibits a related pointwise-versus-population distinction. Across different seeds, arbitrary finite label histories can be prescribed exactly. What is not controlled is the asymptotic history generated by one fixed seed.

Accordingly, the two problems share the broad structural theme

finite or population-level freedom does not supply pointwise asymptotic control.

This is an interpretive parallel only. We do not prove that Chang’s return-balance problem is equivalent to the moving-anchor problem, that one implies the other, or that resolving Open Problem G would resolve EOC.

G. Relation to Chang’s original balance problem

The mod-32 observable used in §5.4 is motivated by Chang’s burst-ending analysis [17]. Within the dominant $n \equiv 1 \pmod{8}$ burst-ending channel, Chang distinguishes the two classes

$$9 \pmod{32} \quad \text{and} \quad 25 \pmod{32},$$

corresponding respectively to a gap of length at least two and a unit gap [17, §5]. His companion paper develops the broader structural setting in which this one-bit obstruction arises [18].

The valuation-word identities

$$m_j \equiv 9 \pmod{16} \iff (d_j, d_{j+1}) = (2, 1)$$

and, at such an event,

$$m_j \equiv 25 \pmod{32} \iff d_{j+2} \geq 2,$$

together with the canonical finite-history construction, are derived in the present paper rather than attributed to Chang.

Chang's stated balance requirement should also be distinguished from the stronger diagnostic formulation adopted in Open Problem G. His reduction asks for sufficiently small orbit-level imbalance between the two relevant mod-32 classes, with the allowable deviation from $1/2$ controlled by his block total-variation budget [17].

The exact limiting statement

$$\frac{1}{K} \#\{t < K : Y_t = 1\} \longrightarrow \frac{1}{2}$$

is a stronger formulation considered here; it is not attributed to Chang.

Theorem 5.14 proves neither condition.

H. Existence bounds versus EOC anti-concentration

The finite-history theorem and the EOC realizer problem use the same exact congruence in opposite directions.

For a canonical history y of length K , Theorem 5.14 gives

$$1 \leq r(D(y)) < 2^{4K + \#\{t: y_t = 1\} + 1}.$$

This is an existence-oriented upper bound for a specially constructed family of words.

By contrast, the residue-axis form of EOC seeks a uniform lower bound of the type

$$r(D) \geq 2^{\varepsilon N - O(1)}$$

for arbitrary relevant c -confined words.

The fixed-anchor results establish such lower-bound behavior only in periodic and eventually periodic symbolic sectors [15]; the genuinely irregular moving-anchor sector remains open.

Consequently,

finite constructive realizability $\neq$ least-realizer anti-concentration.

The finite construction therefore neither proves nor supplies an arbitrary-word EOC realizer floor.

I. Canonical histories versus confined EOC words

The canonical construction should also not be identified with the c -confined family relevant to EOC.

For the repeated block $(2, 1, 1)$, corresponding to the all-zero canonical history, the within-block increments are

$$2 - \alpha, \quad 1 - \alpha, \quad 1 - \alpha.$$

The maximum excursion is

$$R_1 = 2 - \alpha \approx 0.4150375,$$

while the net drift per block is

$$4 - 3\alpha < 0.$$

Hence the all-zero canonical family is c -confined whenever

$$c \geq 2 - \alpha,$$

including $c = 1$, but not for

$$0 \leq c < 2 - \alpha.$$

For the repeated block $(2, 1, 2)$, corresponding to the all-one history, the net drift per block is

$$5 - 3\alpha > 0.$$

Consequently, sufficiently long all-one canonical words leave every fixed corridor.

Thus

finite Chang-history universality $\neq$ universality within the c -confined EOC family.

The canonical construction does not resolve or parametrize the genuinely irregular confined-word sector described in Remark 5.12.

J. Final logical boundary

For clarity, the status of the finite return construction can be summarized without introducing any additional cross-references.

Status	Statement
Proved	Every nonempty finite binary history is realizable by a positive odd seed.
Proved	Within the controlled valuation prefix, the prescribed events occur consecutively and at exactly the designated positions.
Proved	The mod-32 label at every prescribed event is exactly the chosen binary label.
Proved	For a history y of length K ,
	$r(D(y)) < 2^{4K + \#\{t: y_t=1\} + 1}.$
Lean verified	The corresponding finite formalization compiles without sorry , admit , or sorryAx .
Not proved	Every infinite binary history is realizable along a single orbit.
Not proved	The event labels have asymptotic frequency $1/2$.
Not proved	Chang's sufficient pointwise balance condition.
Not proved	Any implication from the return-balance problem to EOC.
Not proved	An arbitrary irregular confined-word realizer floor.
Not proved	EOC or the Collatz conjecture.

The significance of Theorem 5.14 is therefore not that it supplies an asymptotic mixing theorem. Rather, it identifies where such a theorem cannot come from: finite arithmetic admissibility imposes no forbidden binary histories.

Any remaining obstruction must involve genuinely long-range behavior along a single orbit. This is consistent with the pointwise-balance obstruction emphasized by Chang [17, 18], while remaining logically distinct from both Chang’s sufficient-balance condition and the EOC moving-anchor problem.

References

- [1] S. Eliahou, The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths, *Discrete Math.* 118 (1993), 45–56.
- [2] O. Rozier, The $3x + 1$ problem: a lower bound hypothesis, *Functiones et Approximatio Commentarii Mathematici* 56 (2017), no. 1, 7–23.
- [3] R. Terras, A stopping time problem on the positive integers, *Acta Arithmetica* 30 (1976), 241–252.
- [4] C. J. Everett, Iteration of the number-theoretic function $f(2n) = n$, $f(2n + 1) = 3n + 2$, *Advances in Mathematics* 25 (1977), 42–45.
- [5] M. V. P. Garcia and F. A. Tal, A note on the generalized $3n + 1$ problem, *Acta Arithmetica* 90 (1999), no. 3, 245–250.
- [6] M. J. Curry, An Explicit Windowed Sparsity Bound for Divergent $3x + 1$ Orbits, with Logarithmic-Floor Exclusion beyond the Harmonic Barrier, Zenodo, 2026. <https://doi.org/10.5281/zenodo.22087163>
- [7] J. C. Lagarias and A. Weiss, The $3x + 1$ problem: two stochastic models, *Annals of Applied Probability* 2 (1992), 229–261.
- [8] A. V. Kontorovich and J. C. Lagarias, Stochastic models for the $3x + 1$ and $5x + 1$ problems and related problems, in: *The Ultimate Challenge: The $3x + 1$ Problem*, Amer. Math. Soc., 2010, 131–188.
- [9] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum of Mathematics, Pi* 10 (2022), e12.
- [10] R. P. Steiner, A theorem on the Syracuse problem, *Proc. Seventh Manitoba Conf. Numer. Math. and Computing*, 1977, 553–559.
- [11] J. Simons and B. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, *Acta Arithmetica* 117 (2005), 51–70.
- [12] C. Hercher, There are no Collatz m -cycles with $m \leq 91$, *Journal of Integer Sequences* 26 (2023), Article 23.3.5.
- [13] D. Barína, Improved verification limit for the convergence of the Collatz conjecture, *The Journal of Supercomputing* 81 (2025), Article 810. <https://doi.org/10.1007/s11227-025-07337-0>

- [14] M. Lothaire, *Algebraic Combinatorics on Words*, Cambridge University Press, 2002. (Sturmian words: Chapter 2, by J. Berstel and P. Séébold.)
- [15] E. De Jesús, Periodic and Eventually Periodic Realizer Floors in the Accelerated Collatz Carry Equation: Fixed 2-Adic Anchors and Binary Periodicity, technical note, 2026. <https://doi.org/10.5281/zenodo.21889748>
- [16] E. De Jesús, The Entropy-Deficit Identity in the Accelerated Collatz Word Model, Zenodo, 2026. <https://doi.org/10.5281/zenodo.21889866>
- [17] E. Y. Chang, A Structural Reduction of the Collatz Conjecture to One-Bit Orbit Mixing, arXiv:2603.25753 [math.DS], 2026.
- [18] E. Y. Chang, Exploring Collatz Dynamics with Human-LLM Collaboration, arXiv:2603.11066v6 [math.DS], 2026.